RxInfer ~ March 7, 2024 ~ Hidden Markov Model

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okay welcome to RX and fur on March 7th 2024 Andrew to you first thank you sure yeah so um last we so we've been following through on looking at the RX and fur examples that they've posted on their site at reactive Bays last week we went over the coin toss example we got to look at some basic Syntax for putting together a model and generating data and then running inference as well as looking at some plots along the way to see the results of the infer function that's in the package so this week we've jumped to uh under basic examples to how to train your hidden markof model example um and here this is getting a little bit closer to active inference now we're looking at things like uh State Transitions and observation to State likelihood mappings um things that are getting a little bit closer to uh what we find say in the textbook and so um I think to start does anyone have any thoughts on uh for anyone who is able to kind of work their way through the code and the expl um explanation on the site anyone have any kind of like thoughts initial Impressions on what um what's going on in there or or otherwise one second just pulling up the code Andrew I was uh can you hear me okay yes yeah I was um uh quite busy so I couldn't really get to the code although I did manage to run it through but I first tried to to uh approach the theoretical part that that aligns with this code and that took me to to uh sanj's book I believe chapter nine uh which you guys agree that this uh must closely maps to chapter nine of San's book okay okay so I don't know was there more discussion on that that we needed to have before my stuff partially observable Markoff decision processes so yeah I don't exactly remember nine but yes that's definitely the closest from sanji's fundamentals book gotcha gotcha yeah there was no um while I think it's great to try try and find overlap between the two yeah there was no requirement uh for this particular project to to cross reference with Sun's work um although I imagine it's super helpful we did have uh I noticed someone posted a question uh in the meeting notes um they noticed that the Roomba example was a little confusing because they called the transition Matrix the a Matrix in the State observation mapping Matrix the B Matrix whereas um inact in matrices in the textbook it seems to be the other way around um and just I'll speak on that and just say yeah there's like as far as notation goes from what I gather um yes that versus like looking at a PDP or even just an hmm in the textbook we have like this is how we're setting up the model and so we have our categorical distributions per usual and deay distributions play a role later um so a lot of things are very similar but that said like um X is being used to denote in this example observations whereas in much of the textbook if we see an X it's usually referring to States um and then yeah with the the A and B matrices just seem to be like flipped right so so our state transition Matrix is denoted by a whereas normally it's in the textbook it's B observation Matrix like mapping observations to uh states where in this example we're mapping the flooring to different rooms in the place where the Roomba is going yeah that's that's our likelihood Matrix which is normally denoted by a in the textbook um Andrew can I have a quick comment on that I'd think coming from the engineering side me being an engineer by training I know in the state space area the standard convention has always be X do = ax + bu U and Y = C CX plus du so I'm not surprised that people get confused by the flip of the A and B matrices um as a fact the um the Matrix that matches from the current state to the next state in engineering has always been a and the Matrix that Maps the um you know the state to the observation has always been C so I can understand that confusion yeah I know that that makes a lot of sense and I'm sure that a lot of the work that's being put into developing RX and fur and so on so many folks are working in like a more engineering oriented background uh it would make a ton of sense to me to just kind of continue using the notation as it's been developed there yeah so I guess for for us or anyone coming for a more active uh background yeah I can see where the confusion would come up uh just one of those little things to track um able to just follow along with the variables as we go yeah and that just the remaining part I noticed from the buyer lab side they seem to almost always use X for the observation which I found uh confusing too in uh you know they being Engineers uh usually Y is used for the observation but they use x but I personally think that comes from their course in um Asian machine learning where they started basically and just have a x variable and then gradually they expand that to to include time as well but I've notic that too yeah yeah for sure I'm so used to myself to seeing like why is being your your outcomes or observation we have that in the the active inference textbook as well okay does anyone else else have any thoughts or or questions about what's going on in here otherwise we might be able to just do like a quick run through of what's going on in the code I think from my perspective because I was reading through so hidden marov models are something that we utilize quite often in genomics um for finding uh different regions on the genome um and a lot of times we utilize and maybe this is a terminology thing as well I'm not sure um it's been a while since I've done a hidden mark off so I had to look up somebody else's example um you have the transition uh matricies the start matrices and the emission matrices and this implementation um is a little bit simpler than some of the stuff that I've seen from like something written in Python and I don't quite understand the differences between the A and the B States in this um and the C and the use the categorical I guess function uh as I'd call it in Julia um for the S not um is there any Clarity on anybody able to provide some clarity on what these things actually are in relation to a more traditional hidden marov model and if you that's what if you scroll down to the model specifications itself the function of the Hidden marov model not the generation of the test data sure so we're looking here yeah so this is where it gets a little bit more confusing for me because the A and the B Matrix that are just a and b Matrix um with an s there I see the constraints are in a probability form that we can use but I'm not entirely sure haven't quite been able to map it back to what it in a traditional hidden marov model is actually happening here sure um is thata s sore Theta is here yeah is that Theta yeah I'm I'm actually not sure myself I've noticed in in other arcs and for like examples and documentation you can actually directly type A Theta into like the code um here too um got to use a special backslash notation uh at least in vs code so gotta yeah here uh as far as I understand it right so we're so we're defining the model we're giving it some kind of um prior specifications about A and B so a being State tostate transitions moving from one room to another B is like the models or the agents the roomba's ability to uh you know come into contact with X data points uh floor here being flooring and then being able to infer I'm in uh s room um and so sore Theta if if that's indeed a Theta I well h i noticed the that's that's giving you your prior isn't it your initial prior that's right this would be the initial state right and so you're setting distribtion yeah and then so it's it's being filled with ones but then I I this yeah it's being filled with threes right so categorical is just is just a function they giving you to make the code easy to fill those those matrices and it's filling it it's filling it with a uniform point you know your uh your sort of initial prior I guess that's the uh is being filled with uh uniform 3s yeah I think that's right and then um so being able to have a categorical distribution and a de using de lay distributions just intuitively it allows you to specify like okay for initial States we know we're looking at States we know there's a total of three rooms and so we end up with this like 3x3 um a bunch of 3x3 matrices that are that are doing the mapping and then because it's categorical we can find ourselves in a spot where we move from distributions to actually specifying a particular um output right otherwise we're just staying in the realm of like having a bunch of 33s everywhere um like earlier on when we're generating data um for my understanding here what this random

Vector is doing for example this is taking like setting up Cate ories where we have three different um potential States and then here we just draw from one of them and set it to a one um so I'm I'm not I might not be I might not be doing the best job of explaining this but it's like actually being able to go from a probability distribution to actually being able to say oh specifically this room or specifically this observation it's what is what's being um countered so yeah I'm I'm 100% sure that's not a s th but it is a s zero that's the way it's rendered in my notebook yeah that seems to be right because whenever I just look at another zero we have this like you know a zero with a dot in the center I guess it's a diamond if I really zoom in similar looks almost identical right yeah seems to yeah it looks like the prior on a the prior is a based on the Matrix I'm not even try to pronounce that um based on what they're writing so the prior there is the a which is there sitting into just ones filling out a 3X3 Matrix with ones um I mean the setting up initial States the standard like the probability of and I'm I'm going to try not to throw out like different like X and uh Pi uh that are more standard to this my applications but the probability of x given y essentially is uh the probability that the hmm is going to follow a path of Y and emits a string X and I'm trying to and that's going to be equal to the product of the emission and the transition uh Mission and transition probabilities like this is something that we utilize a lot and I'm trying to understand where the emission and the transition probabilities are in this and how that probability and and I don't think that this is like a onetoone mapping I'm just trying to understand where these things go in this math yeah definitely I mean from from Reading like the explanation here it looks like whenever we're defining the model and giving it prior over State transitions this is like basically equivalent to um giving it like a uniform prior like we're we're all ones like nothing gets stronger preference like going from living room to living room is a one going from living room to bathroom is a is a one like it doesn't know right versus the reality that I believe we actually Define the data there's like you know one of these is um yeah entirely z0 I think one of the rooms you just cannot actually in reality in terms of like a generative process you can't actually go from um you know one room to another in this instance like a particular room to another particular room so that's the difference here like defining a generative agent where it has no idea at first and then for b um the reason why we have tens and ones is that uh so tens are all along the diagonal I'm not sure why the code example has the second and third rows like um tabbed way over here um but in any case yeah um on the diagonals we have 10 and the idea is um we're going to we're assuming that the model does have a pretty good good sense of mapping the correct flooring right so I I attempted myself to make some notes Here of what yeah and sorry for the you know this probably looks like a lot but it's like upper left for the B Matrix um going the hard flooring is act in reality the hard flooring is mapped to the bed tiles are mapped to the bath carpet is mapped to living like that's the true um mapping there and so along the diagonals these are all actually correct and then all the rest has some noise that's what the ones are if they were zeros then it's as if the it's as if the room the agent would be perfect in knowing like oh no if I see tiles then I'm definitely not in the bedroom um no instead we have ones to know I mean as and also as the question from the notes pointed out like usually in the in figure 4.3 PDP the B Matrix is hidden State at T minus one to Hidden State at T here that's describing the observation of the flooring to the to the mapping of the Hidden State true location like tile in the bathroom so that's acting more like what we would call an a matrix yeah yeah exactly these are it's like your likelihood mappings yeah just mapping States the the observations um Andrew I don't believe there's any significance in the tabbing over of row two and three I actually corrected it in my code I think it's just an artifact of some some tabs or something that's the way I understand it yeah same here I I just corrected it it's entirely just formatting yeah yeah I found that to be the case as well what is the um that that I'll I'm starting to grasp how that all plays out I think that the straints is where I'm maybe losing myself a little bit there um that looks like it's the actual equation that we're constraining the variables to is that right yeah so this is like you know Q denoting like a variational posterior that we're trying to infer and then we're marginalizing out like here this and initial part this is your um joint distribution between the initial State and your States but you're marginalizing out um yeah the the variational posterior over a your state transitions as well as B your um observ your likelihood um Matrix your your observation States mapping and they claim that that is yeah it's a dependency coupling that allows the inference to actually be tracked gotcha like I personally am also trying to kind of wrap my head around that it seems clear that a lot of this kind of modeling we're doing a lot of marginalizing out so that yeah we can render things tractable but I can't say off the top of my head I know exactly how that works um it just seems clear that yeah often we're confining we're defining constraints whenever we build these um so in all of this and I'll have to just kind of noodle over some of these our discussion here to work through kind of the similarities or differences mostly differences between the two's the more probabilistic approach that I understand via the how I learned it uh in this approach um so this is all essentially generating the matrices that you need to go into a hidden marov model um and well really learning it it's essentially the um was the Brum Welch style um that I'm used to um of learning a matrix the example and it doesn't look like I I didn't see this anywhere the example has has anybody tried to actually utilize those matrices then um to predict something that's unknown yeah any know how to do that well it would be it would be relatively straight forward right because I mean you're saying take the uh learned the Learned um matric values and then just do a blind you know generate a new data set based on that certainly you could do it easily if I got your question correctly that's yes yeah so I mean to do it is easy enough because then you just Define in your model you you just copy those matric matric values in and then you it'll it'll generate data set all you want but so that yeah you could definitely do that you could also do experiment to the code yet you could yeah I mean you could that would be straightforward you could also do the experiment where you could say well you know you know what the generative model values are or the generative uh uh system Mo values are you know what the model values are you can run both compare those the results get distribution of the errors you could do all that well and I think the challenge for me right now is that and I've read through most of the basic examples on conceptual level it they focus a lot on using the infer function to essentially learn from the data that it is provided I haven't seen any examples of then taking that learned data and then applying it anybody found that in examples I'll just keep maybe just a question if you've looked into the infer function does the infer function only provide fast real-time inference or can the infer function also include what we call learning kind of yeah that's something I may need to just look into the infer function a bit like does infer capture all is it you know is it like a cognition function and it includ attention learning and the fast inference or are there separate functions that are called called like for kind of within trial and between trial inference where the between between trial inference would be called learning just just just thought but we'll explore okay yeah I've personally not made it too far through the examples but it's like I think in this this Advanced one where you're like developing a a car that's learning how to like it looks like it learns how to use moment and movement to actually like surpass a threshold that it can't initially um it looks like this example involves like over 50 time steps it starts to learn like it it does like policy comparisons and then it learns over time like how to get enough momentum to actually like overcome a threshold versus like earlier on in the example it just I don't there we are like it can't actually reach the goal because this is where it's stuck right I I really like these um these animated uh visualizations but yeah I I haven't looked into the maths yet and the code for this yet though maybe we can move on to this in the in the future to get a sense of like is it actually learning like how is it doing that sequentially through iterations what are what are the variables being finded here to do that or track that yeah I'm I'm sure we'll find essentially question answer my question here uh throughout this course um just kind of interested in once we learn something how do we use it in unknowns um unknowns of the same data schema at which point the infer function develops you at the very least summary statistics that you can do Downstream analysis with from the data with that initial schema then but that's still not yet this like um the kind of next two Vistas are active engagement with the data and then the second one would be flexible or variable data schema but if if you already second one ahead oh sorry I was just going to say because and I think it's more concrete if I kind of explain what I'm trying to think through and how I'm trying to think through it is if I learn um from a a human genome from a human genetic sequence the that this is the and this is just the example the hidden marov model for um a competitive reg trained a hidden Markov model to identify these regions in this genome now if I take that model and move it to chimpanzee which is 99.9% similar to us how do I use that model to understand where their repetitive elements are if that makes sense yeah well in a nonactive like hmme which maybe even something you're using you learn the transition frequencies on the human genome from like let's just say annotated repetitive regions and then um anything that has atcg genome you can then roll out that hmm to its genome um and you're going to get a probabilistic assignment is saying okay under the templating that we learned from the human repetitive regions of these kinds here's How likely these are these regions are to be those repetitive but then if there's another kind of repetitive sequence in another species of course you wouldn't be able to detect it or if if there was a variant it you know so it it's like the confidence the the statistical instrument that was calibrated on the human repetitive elements it would be a secondary question how applicable was it further further away F genetically absolutely and conceptually that's absolutely correct I'm trying to get down to the nitty-gritty of if I were to use this hidden markof and this is just because I understand hidden Markov Model A little bit better than I do some of these other um models um if I if I build the hidden Markoff model with their example on a test data set now how do I take that um that learned test set and then do exactly what I just said on a essentially another Roomba in their example um like this Roomba is in a different house but has this sort of uh exact same behavior at least similar behaviors then now how do I take what I've learned from room number one Matrix that I learned from r one and then how do I plug that into the code to instead of just learning something to actually predicting how the Roomba is going to um operate so in your example would what what's different between cases is it the Roomba or is it the house because it would make a difference I wouldn't be able to answer that one um I'm just thinking from like a purely code perspective how do I use what I've made okay house the

the house is the species and then the the flooring is The genome sequences that we're observing and then the different rooms are are you know genic intergenic repetitive regions intergenic regions are tiled genic regions are carpeted and so so we're we're feeling our way along the sequencing data only be okay tile tile tile carpet carpet carpet inferring what part of the um what room we're in then so the room so you train on that house Human Genome then pick up the rooma move it to a different genome now if that house was like in the same prefabricated neighborhood then all the mappings would be the same so then that mapping that you had what what they call the B Matrix but what um is the mapping between hidden States and observations like that will exactly apply but if you moved it into a house where the bathrooms were carpeted and then like the living room was tiled it would be giving you incorrect information absolutely but that Matrix itself can be can be calibrated on the known and then you just that's the exact artifact that you take out but then you you need to pull out at least one layer to ask like how adequate is that in a new setting absolutely um Andrew would you do me a favor and go to back to how you TR your hidden marov model I'm probably not being very uh clear um if you scroll down to the posterior marginal for a the 3X3 Matrix the essentially the result section of this yeah that yeah scroll up there we go perfect so these are the results that we generate this uh posterior marginal for a uh and posterior marginal for B if you now Andrew if you'll scroll up to the results my understanding is that this results code I'm sorry one more up to the infer yeah this my understanding is that this infer function is essentially building those marginal matrices understanding from the data um I data um what the what it the model needs to learn what the matrices need to understand so this is essentially the Bram Welch um kind of approach to this where you're learning from data what the posterior and or what the marginal uh distribution is for these matrices so that you learn your matrices your um in my terms the or my understanding the emission and the transition Matrix now if we were to then want to use those matrices how do I do that do I if I use the infer function it's just going to learn again right or or am I wrong you know if I if I give a data in there uh instead of I data um what is it going to do is the code going to essentially build a new results model train it um how do I plug in posterior marginal for a and the posterior marginal for B back into the hidden marov model without the hidden marov model having those priors these are the new priors that it's going to have now how do I utilize those matrices in the way that you're talking about Daniel the um moving it moving the Ruma to a new house or to a new genome how do I utilize that hdden Markov model in this way so that I can then actually do what you said moving it to another system I mean is it enough to just do the inference to learn and then literally take the snapshot outcome it's like well we we we chugged all the weather data there's a 80% correlation between cloudy and gray now we've just distilled it down to this summary statistic VAR you know Matrix and now we're going to use that Matrix in another way so it's like just you just take the mapping and then just use that as uh in a in a another part of the program using generative model that that again recapitulates the Roomba um setting if it were another Roomba or maybe um the Roomba was scouting out and learning that relationship in the room through a thousand sensory samples and then you just send the summary Matrix to the real estate agent yeah I mean you you could absolutely there's a lot you can learn from these matrices Alone um the summary statistics absolutely cool wonderful I don't want to moan this I think I got the kind of my understanding thank you guys and also let's maybe explore in in shared repo or however people want to like let's try um ID data J data H data and like what what happens if if we just or just have three different priors and then just do iterate over three options or something just to give a little like flush out the space a little bit more what happens when you go in with a looser prior or what happens when there's you know something else absolutely and I'm working through and I meant to do this earlier this week I just out of time um I'm working through an example um through a uh bioinformatics algorithms an active learning approach fantastic book um but I'm working through some of their examples using python using memor more traditional way essentially this is the book I learned from implementing their examples and then I'm going to try to bring that into Julia into this uh RX and fur and see if I can make that actually happen the same way and I'll share that code um here this this next week yeah nice that sounds very interesting and I I I personally was playing around just with like number of iterations here um I like jumping to like a 100 iterations the the results are not like dramatically different but you you could guess that they're different um so we're just like trying to yeah do typical inference like have the model learn like what's your a matrix what's your B Matrix compare that to because we get to see on the inside because we're we're generating the data ourselves we know exactly what it looks like um it's up here here are your true right like your true A and B matrices we're just trying to approximate that using the model not to be redundant I think I think we're all pretty much on the same page with with that how it works but yeah worth reiterating perhaps does anyone else have any thoughts on the code I mean it feels like we've been progressively making our way through the code actually um we're just a little bit away from the you know the final plots here which I believe pretty seem pretty self-explanatory it's nice getting familiar with the plotting function though like here P1 a plot we're like just putting in the Sate the true

States data that's generated from uh the generate data function we defined earlier we update P1 plot to also include the result State inferences of the model then we have our labels real and inferred and so that's this this is P1 this upper plot real thir um some slight differences here and there but it's is generally accurate P2 is just grabbing from that result from the infer function the free energy calculation over number of iterations so the default code they use iterations equals 20 there so we end up with we get to watch the free energy being minimized over time looks like it generally levels out sometime around maybe the eth or 9th iteration looks rather static from there um on the on the free energy I mean this is the key point that the maximum evidence model is the least surprising one rather than pursuing a simple maximum likelihood or leloy and approximation or any other kind of like direct evidence maximization method free energy principle suggests pursuing variational free energy bounding of surprise that's what the every parameter is being fit with this kind of common approach to a loss function um the Y AIS is arbitrary or kind of like locally contextualized units so it's not like 110 units of of this Roomba free energy is you know 77 units um they're not they're not directly meaningful units which is like why usually uh the gradient or the Delta f is what is discussed and then I think just from this plot like it is monotonically decreasing um it's always going down but then when there's um noisiness or changes in the environment we see plots where like free energy can Spike and also um by analogy to the loss function with neural networks like live stream 51 it's not clear like what it might stabilize at 80 and then like another 50 iterations later is it possible it drops down to another Plateau like that is seen a lot in training other basian models so especially if you're looking at joint distributions of more than one variable it's possible that like it could stabilize and then drop again but in a in a really like in a single you know small Matrix this is probably stabilized and you could confirm that like by doing the exact basing calculation or by calculating surprise and finding like now we're actually near the global surprise Minima but in a joint distribution there might or where there's stochasticity or environmental change the free energy traces are probably a lot more complex than this sure sure and I I suppose also just um more basic thing like just to refer back to what was being discussed like say build up a Roomba stick it in an entirely new room um like if you threw it into a room where all of the observation state mappings were different and so you have like I don't know the bathroom has hardwood floor and the ba uh living room has has tiles and so on I imagine like even though we've trained this Roomba well enough we end up with our P our posteriors we took those plugged them into um being used as the priors for our Roomba now like you know we trained it to go through this house you put it in a new house like you would see free energy probably start to shoot way up because it's saying you know entirely new um observation State Mings and another funny thing with with um is like so you put it into a different house and then all you saw was the free energy trace and then again you don't know what the numbers on the y- axis mean so it looks like it's stay stabilized you think okay we're doing as well as we can in this new house but the model could be doing terribly it's just out of stationary absolutely inadequate estimation level but it's not getting better or worse from how bad it is which similar to to like a linear regression putting it on a new data set it will find the line of best fit and it and but you have to do diagnostics to find out how how adequate is that line of best fit like What proportion of variance explained okay it went from 140 arbitrary units to 80 arbitrary units but did it go from explaining none to all of the statistical variants or did it go from explaining like two to 3% of the statistical variance and then that's the you know difference between statistically significant and conversationally meaningful yeah for sure so I guess it's so the a takeaway is to just bear in mind like these are arbitrary units and their only usefulness is like kind of relative to the model itself like okay I don't know exactly what 140 means I don't know exactly what 80 means I don't know what that means in this house versus another what I do know is that like there's optimization going on here over time is it work cost function like this is decreasing and that's kind of the the meaningful aspect here that we we can take away well and if you were to and my understanding is if you you know you apply this to another data set as you say then you know you might see it flatten out and you know kind of that the the model is kind of reaching its limit with how it's currently designed in the priors that you're giving giving it and it might be doing terribly it might be doing great um but at least you know where the limit of the model is at that point yeah actually my own question here because I've mainly just spent time trying to kind of digest what's going on in the um kudal paper um where we had previously had a live stream think was live stream 55 uh where they're going over uh direct policy in France and it's using the RX and for package here they spent a lot of time talking about kind of the the more recent history of different derivations of free energy and so we have you know we have expected free energy generalized uh free energy we have be free energy um is it the case that the RX and fur package here like setting you know we set free energy equals true in the fur package so that's what's being used for optimization um is it Beth free energy like it's it seems to be the case that you can definitely use it in RX but I don't know if that is the default um like is is is that what we're doing whenever we run this particular code the How to Train Your Mark model um that's probably the case ber I remember when we did 55 if you looked into the cffg and the be or the beroun but yeah I'm not know how but B do you remember any of this uh sorry I wasn't following I was putting in the the posteriors for the transition matrixes and seeing like put them them in as the marginal which you start and see if the the free energy converges more quickly and it seems to do that so I didn't follow what you guys were saying okay all all good just just like which free energy flavor is being used here generalized variational be I think they mentioned bet somewhere in one of the the code [Music] BS because they do apply the the message passing on the the factor cph that's also why you need to add the constraint the the yeah the factorization constraint uh but I think this is just bad yeah also is it um the case that for a simple like if you're just doing a a sing single variable it might not matter but if you were doing like if you were doing um iterative optimization or if you were looking at graphs with Cycles then the the the Beth matters more yeah the generalized the generalized free energy I I this is just speculative I think the generalized free energy is a generalization of variational and expected so generalized free energy captures the kind of real time and the prospective Elements which is what enables it to be used for the um direct policy inference whereas the the bet is an approximation factorization um technique but the generalized free energy is is a generalization in principle across VF and and efe possibly even the fef all those kind of alternative future free energies and I think the bet that free energy is more in the um meanfield space of just like what are what are ways you can take a node and then approximate its free energy um given like for example Cycles in the factor graph but any any or multiple aspects that we can like pull out to specific questions then and and also especially if it connects back to like the documentation if we're just like oh it wasn't clear in this part what which free energy was being used or like let's just ask bias lob get their put improve documentation yeah definitely all right does anyone else have any other thoughts anything they've done um as well as I noticed we only have about five or six minutes left here um so also any kind of discussion on maybe what we want to do next or actually no sorry we do have a looks like next week we are going to hop to that mountain car example which is under the advanced examples this one this one should be fun um live stream8 very early with Alex shance is also discussing the mountain car problem um I don't even know if that is using pmdp we can look back to the code it's either using PDP or a non pmdp python code form um especially if the um kind of the the benchmarking and the set uping of the mountain car is consistent with the with like the open AI gym formulation and it might be those are like some of the best contact points to to you know share with people who are familiar with those gym and testing environments again especially if we can show some of these other Diagnostics of the model like speed to convergence computational resources interpretability modularity so on not just training like one big monolithic neural network to to to do a better car it's like we can inspect the policies and all these different things if if if someone or multiple people want to prepare to like present it or to run the code just add your name or like write comments in that section but thank you Andrew this was this was awesome and very helpful with hmm awesome yeah for sure what could you tell me again the name of the what was the reference number for that um live mention I want to check 008 008 just like that yeah and that was in the that was in the uh it's called scaling active inference and so that was kind of earlier on when it was like well we could replace the Matrix with the neural network and it's was like well this helps us address large data sets but does it kind of remove some of the analytical Elegance or is it really active inference all those kinds of questions just looking ahead p uh past the mountain card I think the mountain card that sounds really interesting I'm sure it's That's a classic might take us two weeks to get through it I don't know but looking past that next one are there any examples that that would be accessible to us that move more along the concept of uh self-organization the potential of uh you know how how the you know kind of the full um marov blanket and individual agents that have some self-organizing properties are there examples that would be accessible to us that we might be able to look at look at going forward so like let's ask um Marco and and bias loveb like even like before next week or next week let's say hey we did you know we did coin flip we did hmm we're in Mountain car for one or two weeks and then we want to look to a self-organizing example and see if they know cuz obviously the best the other thing that comes to mind though is the um controversial but impactful 2013 paper Life as We Know It by friston which did kind of speak to that self-organizing element now obviously that was not written in RX and fur but we could look at the paper see if the codee's there email Carl get the matlb code for for Life as we know it and then you know do like Life as we know it in 2024 in RX and fur because that that that um paper has and continues to attract a lot of like the philosophical discussion and then that kind of frustrates um some of the uh people on social medias because it's like no you're looking at a version that's 11 years old and they're like well you should have told me if the theory changed and then they're like we did we wrote a paper about it there's also a paper and I and I'm sure uh you'll remember the authors immediately but there was a a paper that was done really about uh social you really social media organization and Echo chamber Dynamics and I think that was in pi MDB yeah epistemic communities yes epistemic communities and that would be possibly another way to go yeah that that that would yes yeah so maybe one or two weeks with mountain car um meanwhile we will be kind of continuing to sketch out our own projects plus emailing people we'll email bios laab um check in with them ask about self organization and then we can um on the kind of like reproducibility and uh wavelength we can do there's the textbook examples and then there's epistemic communities in Python and Life as we know it in matlb and I mean we can just engage with authors of non RX and fur packages and just say as part of our learning we're looking for we're looking to translate your package into or your your script or your simulation into RX and fur could you just double check and send us the total versions of the code and then like we could go from there and then that that could be awesome and start to bring all these different and I mean that's one reason why the models are kind of hard to compare I mean how do you compare a 2013 model of self-organization in mat lab to paper you know in Python but not pmdp to 2023 in pmdp if for it's like they differ in their implementation in their system of interest in the phenomena that are being measured so there's a kind of like highlevel consilience under the free energy principle umbrella but then yet the models are so disperate but when we bring them into RX and fur and we start to look at how the um the the um recombinations and the composability of the the app models and stuff like that a lot of those pieces might start to click better okay I'm going to stop the recording thank you Andrew thank you everybody thanks